HARD · ADVANCED MATH

Advanced Math

04

10 questions · ~15 min

SAT QUESTION BANK

2026-05-29

Q1 GRID-IN ADVANCED MATH

Function f is defined by $f x = -a^x + b$, where a and b are constants. In the xy-plane, the graph of $y = f x - 12$ has a y-intercept at $0, -\frac{75}{7}$. The product of a and b is $\frac{320}{7}$. What is the value of a?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q2

GRID-IN ADVANCED MATH

$$y + k = x + 26$$

$$y - k = x^2 - 5x$$

In the given system of equations, k is a constant. The system has exactly one distinct real solution. What is the value of k?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q3 GRID-IN ADVANCED MATH

If $4^{8c} = \sqrt[3]{4^7}$, what is the value of c?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q4

GRID-IN ADVANCED MATH

The function f is defined by $f x = x - 6x - 2x + 6$. In the xy-plane, the graph of $y = g x$ is the result of translating the graph of $y = f x$ up 4 units. What is the value of $g 0$?

STUDENT-PRODUCED RESPONSE

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q5

GRID-IN ADVANCED MATH

$$xx + 1 - 56 = 4xx - 7$$

What is the sum of the solutions to the given equation?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q6

If a and c are positive numbers, which of the following is equivalent to $\sqrt{(a+c)^3} \cdot \sqrt{a+c}$?

A. $a+c$

B. a^2+c^2

C. $a^2+2ac+c^2$

D. a^2c^2

Q7 GRID-IN ADVANCED MATH

$$f x = x - 2x + 15$$

The function f is defined by the given equation. For what value of x does $f x$ reach its minimum?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q8 GRID-IN ADVANCED MATH

$$\sqrt{x - 2^2} = \sqrt{3x + 34}$$

What is the smallest solution to the given equation?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q9

GRID-IN ADVANCED MATH

One of the factors of $2x^3 + 42x^2 + 208x$ is $x + b$, where b is a positive constant. What is the smallest possible value of b?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q10 ADVANCED MATH

The function f is defined by $fx = a2.2^x + 2.2^b$, where a and b are integer constants and $0 < a < b$. The functions g and h are equivalent to function f, where k and m are constants. Which of the following equations displays the y-coordinate of the y-intercept of the graph of $y = fx$ in the xy-plane as a constant or coefficient?

I. $gx = a2.2^x + k$

II. $hx = a2.2^x + m$

A. I only

B. II only

C. I and II

D. Neither I nor II